

José Teodosio Lorente Vallecillos

AI & ML ENGINEER | HEALTHCARE AI | LLM AGENTS | DATA ENGINEERING

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Staff-level AI and Software Engineer with experience building and deploying production-grade AI systems in health-care and data-intensive environments. Strong in Python backend development, LLM-based agent workflows, retrieval-augmented generation (RAG), and secure data pipelines. Proven end-to-end ownership from system architecture and data integration to deployment, monitoring, and iteration with real users under reliability, latency, and privacy constraints.

Experience

Devera, Head of AI & Data Engineering | Remote | Spain Sep 2025 - Present

- Sole engineer who built the entire technical vertical from scratch (data pipelines, AI infrastructure, document generation, cloud, microservices), rescuing the company from near-shutdown to active M&A discussions.
- Designed and implemented end-to-end LCA carbon footprint estimation platform: client data ingestion, processing, curation, AI-driven analysis, and automated report generation.
- Architected and deployed RAG systems, local cloud AI models, and orchestration services, owning the full lifecycle from design to production monitoring.
- Enabled expansion from a single city (Valencia) to operations across 7 European countries through scalable, self-built infrastructure.

Remote Care Partners, AI & ML Engineer | Remote | Florida, USA Sep 2024 - Sep 2025

- Built AI-driven patient monitoring systems using time-series physiological data, enabling clinician alerts and triage workflows.
- Developed quantized CNN for secure face authentication, optimized for low-power NPU inference on healthcare hubs.
- Integrated real-time biosignal data (ECG, SpO2, HRV) into secure cloud pipelines for continuous patient monitoring.
- Deployed predictive models for hospitalization risk, achieving 92% F1-score and reducing false positives by 30%.

Remote Care Partners, AI & ML Engineer Intern | Remote | Florida, USA Mar 2024 - Aug 2024

- Implemented end-to-end ML pipelines for healthcare data: ingestion, feature engineering, training, and deployment.
- Developed weather-aware health recommendation feature integrated into patient-facing healthcare application.
- Reduced data processing latency by 45% through optimized ingestion and secure storage for real-time analytics.

Skills

Languages	Python, C++, SQL, JavaScript, TypeScript, Dart, Bash
Backend & APIs	FastAPI, Flask, REST APIs, Async IO, Authentication, Integration patterns
LLMs & Agents	OpenAI APIs, Hugging Face, RAG, Vector search, LangChain, LangGraph, Agent orchestration
AI/ML	PyTorch, TensorFlow, Scikit-learn, Model optimization, Quantization, Deep Learning
Data Engineering	Pandas, NumPy, PostgreSQL, MongoDB, ETL/ELT, Data modeling, Secure data pipelines
MLOps/DevOps	Docker, Git, CI/CD, Linux, Kubernetes, AWS/Azure

Projects

GAIA Benchmark AI Agent – Demo

Agentic AI & LLM Systems

- Built agentic RAG application using LangChain, OpenAI APIs, and vector retrieval for complex multi-step queries.
- Achieved 90% score on GAIA benchmark, ranking top performer with robust guardrails and context management.
- Engineered multi-tool agent architecture with document retrieval, web search, and reasoning capabilities.

AR System with AI Assistant (Bachelor’s Thesis - 9.2/10)

AI & Augmented Reality

- Developed real-time AR system integrating NLP, CNN-based authentication, and gesture control for hands-free interaction.
- Built low-latency server-client streaming with multi-threaded GPU/CPU optimization using OpenCV, PyTorch, and Keras.

Autonomous Sumo Robot (Competition Finalist)

Robotics & Embedded Systems

- Designed autonomous robot with Arduino, ultrasonic sensors, IR edge detection, and custom strategy logic in embedded C++.

Education

MSc in Artificial Intelligence and Data Engineering (In Progress – GPA: 9.5/10)

BSc in Computer Science & Computer Engineering

ETSIIIT University of Granada | Granada, Spain – **Ranked 52nd Worldwide in AI** (U.S. News 2025)

- Bachelor Thesis: Low-cost AR system with CV, NLP, and multimodal interaction (Grade: 9.2/10)

Certificates

2025	AI Agents Course - Top 500 Worldwide (GAIA Benchmark) , Hugging Face - 90% score; pass threshold is 30%. Officially ranked on HF leaderboard
2024	Data Engineering Bootcamp - Distinction (Top 0.019% of 31,000) , Dataexpert.io - Perfect Score
2024	Microsoft Azure AI Fundamentals (AI-900) , Microsoft Learn - Perfect Score

Languages

English	C1 Cambridge
German	Basic proficiency
Spanish	Native proficiency
Chinese	HSK 1 Confucius Institute